

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Secure login for administrators and disaster management officials to access prediction services.	High
2	Authorization levels	Role-based access for administrators, meteorologists, and disaster management authorities.	High
3	External interfaces	Integration with weather datasets, CSV files, and IBM Cloud deployment services.	High
		<i>services the system must connect to]</i>	

4	Transactions processing	Process weather input data and generate flood risk predictions in real time.	High
5	Reporting	Generate flood prediction reports, dashboards, and risk classifications.	Medium
6	Business rules, etc.	Predict flood occurrence using machine learning models based on weather parameters.	High
7	Compliance to laws or regulations	Follow data privacy and cybersecurity guidelines for handling user and weather data.	Medium
8	<i>Other</i>	Maintainability and portability for future enhancements.	Deployable and maintainable on IBM Cloud

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate predictions immediately after data entry.	Prediction displayed in < 3 seconds
2	Scalability	Handle increasing datasets and multiple prediction requests.	Supports multiple users simultaneously
3	Security & Data Privacy	Protect credentials and stored datasets.	Secure authentication and protected storage
4	Reliability & Availability	Ensure availability during emergency situations.	At least 99% system availability
5	Usability & Accessibility	Provide a simple and user-friendly dashboard.	Easy-to-use interface for authorities
6	<i>Other</i>	Maintainability and portability for future enhancements.	Deployable and maintainable on IBM Cloud